Pre-analysis plan: #ManyDaughters: Impact of Having Daughters on Attitudes, Preferences, and Behaviors*

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1 Introduction

In this project, we study the causal impact of having daughters on individual attitudes and behaviors related to gender equality and political preferences. The study will leverage data from the German Socio-Economic Panel (SOEP-Core), covering the years 2000 to 2020. The hypotheses under investigation were specified by the coordinators of the ManyDaughters project (2025), and span attitudinal, behavioral, and ideological dimensions: from support for gender equality to intra-household dynamics and political leaning.

The identification strategy relies on the assumption that the probability of having a daughter is as good as randomly assigned conditional on parity. This allows us to interpret differences between individuals with and without daughters as causal. This document outlines the planned variable definitions, sample restrictions, model specifications, and inferential procedures to be applied once the data become available.

2 Study design

2.1 Hypotheses

- **H1:** Having daughters leads to stronger support for gender equality (as measured by attitudes).
- **H2:** Having daughters leads to more left-leaning political preferences.
- **H3:** Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).
- **H4:** Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

2.2 Inclusion and Exclusion Criteria for the Sample

The analysis will be conducted using data from the SOEP-Core dataset (waves 2000–2020). In particular, the available datasets consist of *pl*, *hl*, *biol*, *biobirth*, *bioparen*, *pgen*, *hgen*, and *ppathl*. The sample will be restricted to respondents who meet the following inclusion criteria:

- Respondents must have at least one child.
- Information on the gender of children must be available.
- For each hypothesis, we select the latest survey year in which the respective outcome variables have at least 3,000 non-missing observations. Accordingly, we use data from 2016 for Hypotheses 1 and 4, from 2020 for Hypothesis 2, and from 2019 for Hypothesis 3.
- Respondents must have valid observations for at least one of the outcome variables of the hypotheses.
- Respondents must be no older than 60 years of age.
- Respondents must live in the same household as their partner.

2.3 Variables

The following lists provide detailed information on the variables used to define the sample, the core outcome measures relevant to our hypotheses, and the covariates included in our analyses. For each variable, we specify its name and describe how it will be constructed based on the underlying datasets once available.

- **Key variables to build the data set**
  - Indicator whether a respondent has a child: We create a binary indicator that is equal to one if the birth date of the first child (combination of *kidgeb*01 and *kidmon*01 in dataset biobirth) of a respondent (*pid*) was earlier than the survey date (combination of *iyear* and *imonth* in dataset pl), and zero otherwise.
  - Indicator whether a respondent has a daughter: We create a binary indicator that is equal to one if the gender of any of the first 19 children (*kidsex*01 to *kidsex*19 in dataset biobirth, 19 is the maximum number of children listed) for a respondent (*pid*) is equal to 2 (daughter) and the birth date of the respective child is earlier than the survey date, and zero otherwise.

- Age of the respondent: We calculate the approximate age of a respondent by using non-missing information about their birth year (*gebjahr* in dataset ppathl) and substract it from the survey year (*syear*).
- Indicator whether a respondent lives in the same household as their partner: We create a binary indicator that is equal to one if a respondent lives in the same household as their partner (*partner* equal to 1 or 2 in dataset ppathl) in the given survey year (*syear*), and zero otherwise.

- **Outcome variables** The coordinators of ManyDaughters project (2025) have prescribed evaluating one outcome per hypothesis. Our outcomes are as follows:

  - **H1 and H4: gender equality attitudes**

  - Gender equality attitudes: We take the respondents' gender equality attitudes from 5 items (listed below) of the dataset pl. Each of these items is answered on a seven-point Likert scale ranging from 0 (completely disagree) to 7 (completely agree). For the sample of subjects with responses of 0 to 7 in all five items, we construct a gender equality attitudes index, using principal component factor analysis retaining the first factor. The index is then standard normalized by subtracting the sample mean of the index and dividing by the sample standard deviation of the index. It provides gender equality attitudes in standard deviations.
    - $plj0621$: Having a job is the best way for a woman to be independent.
    - $plj0622$: Even a married woman should have a paid job so that she can be financially independent.
    - $plj0623$: If a woman earns more money than her partner, this inevitably leads to problems.
    - $plj0624$: For parents, vocational training or higher education for their sons should be more important than vocational training or higher education for their daughters.
    - $plj0625$: At home, the husband should have the final say.

  - **H2: political preferences**

- Political preferences: We take the political preference of a respondent $plh0004$ from the dataset pl. It provides political preference on a eleven-point Likert scale ranging from 0 (completely left-leaning) to 10 (completely right-leaning).

- **H3: gender equality behavior/outcome in household**
  For the following outcome, we link each respondent to their partner in the same household using *parid* in dataset ppathl.
  - Share of female care and housework hours in the partnership: We take for a respondent and their partner the past week's care work hours during week days $pli0043_h$, Saturday's $pli0012_h$ and Sunday's $pli0016_h$, and the housework hours during week days $pli0044_h$, Saturday's $pli0019_h$ and Sunday's $pli0022_h$ from the dataset pl. If all six values are non-missing for an individual, we calculate the total care and housework hours in the past week for a respondent and their partner. If then a respondent's and their partner's total care and house work are both non-missing, we divide the total care and house work hours of the female by the sum of the female and male total care and house work hours. If both the respondent and the partner have zero care and house work hours, the share of female care and house work hours in the partnership is considered to be 0.5.

- **Covariates**
  - Household income: We connect respondents to their households using *hid* and take non-missing information on households' monthly net income ($hlc0005_h$ in dataset hl).
  - Education: We take the years of education and training *pgbilzeit* of a respondent *pid* from the dataset pgen.
  - Age: We calculate the approximate age of a respondent by using non-missing information about their birth year (*gebjahr* in dataset ppathl) in any year and substract it from the survey year (*syear*).
  - Gender: We take a binary indicator on a respondent's gender (*sex* in dataset ppathl).
  - Age of oldest child: We calculate the approximate age of the oldest child of the respondent by taking their respective birth year ($kidgeb1$-$kidgeb19$ in

dataset biobirth) and substracting is from the survey year (*syear* in dataset pl).

  - Number of children: We take the number of children *sumkids* of the respondent from the dataset biobirth and add indicators for having two and three or more children.

- **Additional balance variables**

  - Composition of children measured by the number of sons of the respondent using the variables kidsex01 to kidsex19 in the dataset biobirth.

3 Analysis plan

3.1 Core specification

For Hypothesis 1 to 3, we will generally estimate treatment effects on our outcomes of interest, comparing subjects with a daughter to those with children but no daughter. That is,

$$Y = \alpha + T\theta + X\beta + \varepsilon, \tag{1}$$

where Y is an outcome of interest, T is a treatment indicator of having a daughter, θ our estimand, and X a set of covariates, which includes a linear term of household's monthly net income, a linear term of the respondent's years of education, linear and squared terms of the respondent's age, a dummy indicating the gender of the respondent, linear and squared terms of the age of the first child, and dummy indicators for having 2 and 3 or more children. In the case of missingness in covariates, we impute zero values and use the missingness-indicator method, as described in Zhao and Ding (2022). We will report heteroskedasticity-robust standard errors. Additionally, we will report the extent to which the covariates and the additional balance variable of number of sons are balanced across treatment conditions.

For Hypothesis 4, we additionally interact the treatment with the gender of the parent. That is,

$$Y = \alpha + T\times D\delta + T\theta + D\gamma + X\beta + \varepsilon, \tag{2}$$

where D represents a dummy for gender with 1 equaling male and δ the interaction effect of interest.

3.2 Interpretation of the Results

We report the results of two-sided t-tests. We interpret whether a test result is in line with the hypothesis if we reject the null hypothesis at the five percent significance level. We do not formally adjust for multiple hypotheses testing.

References

ManyDaughters project. 2025. "#ManyDaughters project." URL: https://www.manydaughters.com/. Accessed: 23-06-2025.

Zhao, Anqi, and Peng Ding. 2022. "To Adjust or not to Adjust? Estimating the Average Treatment Effect in Randomized Experiments with Missing Covariates." *Journal of the American Statistical Association*, forthcoming.